

Previously we proved the Cauchy-Schwartz Equality in complete generality using the construction of a Geometric Algebra. In the next few pages we will give a multiplication table for the basis elements of 3 different planes.

These three examples are equipped with the quadratic forms $B(\mathbf{x}) = x_1^2 + x_2^2$, $G(\mathbf{x}) = 2xy$, and $R(\mathbf{x}) = x_1^2 - x_2^2$. Vector spaces equipped with these quadratic forms are called the **blue**, **green**, and **red** planes respectively. In 2008 Norman J. Wildberger initiated the study of **chromogeometry**, which seeks to understand the ways that these three quadratic forms interact with each other. An important first observation is the identity $G(\mathbf{x})^2 + R(\mathbf{x})^2 = B(\mathbf{x})^2$, but to say much more would require a detailed discussion of orthocenters, circumcircles, and complex numbers. Before that can happen, we should get comfortable with doing arithmetic in these contexts.

Example 1 The blue plane is what most people are thinking of when they describe euclidean geometry. It is the GA generated by the basis vectors $\mathbf{b}_1, \mathbf{b}_2$ subject to the relations $\mathbf{b}_1^2 = \mathbf{b}_2^2 = 1$ and $(\mathbf{b}_1 + \mathbf{b}_2)^2 = 2$, which imply $\mathbf{b}_1\mathbf{b}_2 + \mathbf{b}_2\mathbf{b}_1 = 0$ they are orthonormal as well as $\mathbf{b}_1\mathbf{b}_2 = -\mathbf{b}_2\mathbf{b}_1 = \mathbf{b}_1 \wedge \mathbf{b}_2$ our basis vectors anti commute. Over a linearly ordered field blue geometry is positive definite according to the cauchy-schwartz inequality, and there are no visible null vectors whenever our field does not contain a square root of -1 . A general multivector in the blue plane looks like $a + b\mathbf{b}_1 + c\mathbf{b}_2 + d\mathbf{b}_1\mathbf{b}_2$. We could write down a formula for the product of two general multivectors, but it is a little tedious to do out of nowhere, and not really worth the effort. Instead it is much easier to figure out how basis elements multiply and then look at useful special cases. The multiplication table for the standard basis of multivectors is

Blue	1	$\mathbf{b}_1$	$\mathbf{b}_2$	$\mathbf{b}_1\mathbf{b}_2$
1	1	$\mathbf{b}_1$	$\mathbf{b}_2$	$\mathbf{b}_1\mathbf{b}_2$
$\mathbf{b}_1$	$\mathbf{b}_1$	1	$\mathbf{b}_1\mathbf{b}_2$	$\mathbf{b}_2$
$\mathbf{b}_2$	$\mathbf{b}_2$	$-\mathbf{b}_1\mathbf{b}_2$	1	$-\mathbf{b}_1$
$\mathbf{b}_1\mathbf{b}_2$	$\mathbf{b}_1\mathbf{b}_2$	$-\mathbf{b}_2$	$\mathbf{b}_1$	-1

Notice that the even sub algebra of terms generated by scalars and bivectors is isomorphic to the complex numbers $\langle 1, \mathbf{b}_1\mathbf{b}_2 \rangle \simeq \mathbb{C}$. For vectors $\mathbf{a}_i = x_i\mathbf{b}_1 + y_i\mathbf{b}_2$ and scalar plus bivectors $\mathbf{B}_i = u_i + v_i\mathbf{b}_1\mathbf{b}_2$ we have that

$$\mathbf{a}_1\mathbf{a}_2 = x_1x_2 + y_1y_2 + (x_1y_2 - x_2y_1)\mathbf{b}_1\mathbf{b}_2,$$

$$\mathbf{a}_1\mathbf{B}_2 = (x_1u_2 + y_1v_2)\mathbf{b}_1 + (y_1u_2 - x_1v_2)\mathbf{b}_2,$$

$$\mathbf{B}_1\mathbf{a}_2 = (u_1x_2 - v_1y_2)\mathbf{b}_1 + (v_1x_2 + u_1y_2)\mathbf{b}_2,$$

$$\mathbf{B}_1\mathbf{B}_2 = u_1u_2 - v_1v_2 + (u_1v_2 + u_2v_1)\mathbf{b}_1\mathbf{b}_2.$$

These expressions have significant geometric meaning, and aid greatly in the study of isometries of the plane. To make that easier to see, we can use polar coordinates. Define the

magnitudes of a vector as $|\mathbf{a}| = \sqrt{\mathbf{a}^2}$, and a scalar + bivector to be $|\mathbf{B}| = \sqrt{\mathbf{B}\mathbf{B}^\dagger}$ where $\mathbf{B}^\dagger$ is the scalar plus bivector obtained from writting $\mathbf{B}$ backwards. Given two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, their blue directed angle is defined as

$$\theta_B(\mathbf{v}_1, \mathbf{v}_2) = \int_{\gamma} \frac{-ydx + xdy}{x^2 + y^2} \text{mod} 2\pi$$

for γ an oriented path starting at the tip of $\mathbf{v}_1$ and ending at the tip of $\mathbf{v}_2$, which never passes through the origin. The reader is invited to check that for vectors $\mathbf{v}(\theta) = \cos(\theta)\mathbf{b}_1 + \sin(\theta)\mathbf{b}_2$ that $\theta_B(\mathbf{v}(\phi_1), \mathbf{v}(\phi_2)) = \phi_1 - \phi_2$. The $\text{mod} 2\pi$ in the definition of a blue directed angle comes from the fact that the first de Rahm cohomology group of the punctured plane is nontrivial.

For brevity, let $a(\theta)$ be the directed angle that $\mathbf{a}$ makes with $\mathbf{b}_1$, and $B(\theta)$ be the angle that $\mathbf{B}$ makes with 1, suitably defined. This allows us to write $\mathbf{v} = |\mathbf{v}|[\cos(v(\theta))\mathbf{b}_1 + \sin(v(\theta))\mathbf{b}_2]$ as well as $\mathbf{B} = |\mathbf{B}|[\cos(B(\theta) + \sin(B(\theta))\mathbf{b}_1\mathbf{b}_2)]$, and we have

$$\mathbf{a}_1\mathbf{a}_2 = |\mathbf{a}_1||\mathbf{a}_2|[\cos(a_2(\theta) - a_1(\theta)) + \sin(a_2(\theta) - a_1(\theta))\mathbf{b}_1\mathbf{b}_2],$$

$$\mathbf{a}_1\mathbf{B}_2 = |\mathbf{a}_1||\mathbf{B}_2|[\cos(a_1(\theta) - B_2(\theta))\mathbf{b}_1 + \sin(a_1(\theta) - B_2(\theta))\mathbf{b}_2],$$

$$\mathbf{B}_1\mathbf{a}_2 = |\mathbf{a}_1||\mathbf{B}_2|[\cos(B_1(\theta) + a_2(\theta))\mathbf{b}_1 + \sin(B_1(\theta) + a_2(\theta))\mathbf{b}_2],$$

$$\mathbf{B}_1\mathbf{B}_2 = |\mathbf{B}_1||\mathbf{B}_2|[\cos(B_1(\theta) + B_2(\theta)) + \sin(B_1(\theta) + B_2(\theta))\mathbf{b}_1\mathbf{b}_2].$$

We can conclude from these formulas that $e^{\theta(\mathbf{b}_1 \wedge \mathbf{b}_2)}\mathbf{v}(x)$ rotates $\mathbf{v}(x)$ counterclockwise by θ radians, and $\mathbf{v}(x)e^{\theta(\mathbf{b}_1 \wedge \mathbf{b}_2)}$ rotates $\mathbf{v}(x)$ clockwise by θ radians.

In order to study functions and curves in the blue plane we introduce the vector differential operator $\nabla_B = \mathbf{b}_1 \frac{\partial}{\partial x} + \mathbf{b}_2 \frac{\partial}{\partial y}$ called the **blue gradient**. Given a differentiable scalar valued function $f(x, y) = f(\mathbf{x})$ called a **potential** then we can write the tangent line at $\mathbf{r}$ as $[\nabla_B f(\mathbf{r})]^T B(\mathbf{x} - \mathbf{r}) = 0$ where $\mathbf{x} = x\mathbf{b}_1 + y\mathbf{b}_2$ is a vector of variables and

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is the metric tensor of our quadratic form. This implies that the blue gradient of a potential is blue perpendicular to tangent lines of the potential.

For a vector valued function $\mathbf{v}(\mathbf{x}) = p(\mathbf{x})\mathbf{b}_1 + q(\mathbf{x})\mathbf{b}_2$ then we have

$$\nabla_B \mathbf{v}(\mathbf{x}) = \frac{\partial p}{\partial x} + \frac{\partial q}{\partial y} + \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y} \right) \mathbf{b}_1\mathbf{b}_2$$

which is the blue divergence of the vector function plus its curl times the unit bivector.

Likewise, in order to understand the effect of the gradient on a scalar plus bivector (complex) valued function, we need to look at a consequence of the chain rule. If a complex expression $u(x, y) + v(x, y)\mathbf{b}_1\mathbf{b}_2$ is a differentiable function of a complex variable $f(z = x + y\mathbf{b}_1\mathbf{b}_2)$ then according to the chain rule $\frac{df}{dx} = \frac{df}{dz} \frac{dz}{dx}$ and $\frac{df}{dy} = \frac{df}{dz} \frac{dz}{dy}$. Solving for $\frac{\partial f}{\partial z}$, setting both expressions

equal to each other and simplifying we get $\frac{df}{dx}\mathbf{b}_1\mathbf{b}_2 = \frac{df}{dy}$. By substituting in our complex expression for $f(z)$ into this equation, and then comparing the scalar and bivector parts we get two relations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ called the **Cauchy-Riemann** equations. Now applying the gradient

$$\nabla_B f(z) = (\mathbf{b}_1 \frac{\partial}{\partial x} + \mathbf{b}_2 \frac{\partial}{\partial y})(u(x, y) + v(x, y)\mathbf{b}_1\mathbf{b}_2) = (\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y})\mathbf{b}_1 + (\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x})\mathbf{b}_2$$

the Cauchy-Riemann equations tell us $\nabla_B f(z) = 0$.

Exercise 1 Prove that if $f(z) = u(x, y) + v(x, y)\mathbf{b}_1\mathbf{b}_2$ then $\nabla_B u \cdot \nabla_B v = 0$. For what conditions on $\mathbf{v}(\mathbf{x})$ is $\nabla_B^2 v = 0$? For what conditions on $f(z)$ is $\nabla_B^2 f = 0$?

Exercise 2 Using the previous exercise and the function $f(z) = z^2$, prove that red $x^2 - y^2 = r$ and green $2xy = g$ circles are perpendicular wherever they intersect. Vectors of the form $k\mathbf{b}_1 \pm ik\mathbf{b}_2$ where $i^2 = -1$ are called **null vectors** in blue geometry because $(k\mathbf{b}_1 \pm ik\mathbf{b}_2)^2 = 0$. Prove that if a red and green circle intersect at a blue null point, then their tangent lines at that null point are equal.

Example 2 The red (hyperbolic) plane is what most people are thinking of when they describe **minkowski geometry**. It is the GA generated by the basis vectors $\mathbf{r}_1, \mathbf{r}_2$ subject to the relations $\mathbf{r}_1^2 = -\mathbf{r}_2^2 = 1$ and $(\mathbf{r}_1 + \mathbf{r}_2)^2 = 0$, which implies $\mathbf{r}_1\mathbf{r}_2 + \mathbf{r}_2\mathbf{r}_1 = 0$ they are (hyperbolic) orthonormal as well as $\mathbf{r}_1\mathbf{r}_2 = -\mathbf{r}_2\mathbf{r}_1 = \mathbf{r}_1 \wedge \mathbf{r}_2$ our basis vectors anti commute. Over a linearly ordered field red geometry is non-definite, because it has a basis vector whose square is positive, and a basis vector whose square is negative. Red geometry always has visible null vectors, of the form $k\mathbf{r}_1 \pm k\mathbf{r}_2$, over any field. The multiplication table for the red plane is given by

Red	1	$\mathbf{r}_1$	$\mathbf{r}_2$	$\mathbf{r}_1\mathbf{r}_2$
1	1	$\mathbf{r}_1$	$\mathbf{r}_2$	$\mathbf{r}_1\mathbf{r}_2$
$\mathbf{r}_1$	$\mathbf{r}_1$	1	$\mathbf{r}_1\mathbf{r}_2$	$\mathbf{r}_2$
$\mathbf{r}_2$	$\mathbf{r}_2$	$-\mathbf{r}_1\mathbf{r}_2$	-1	$\mathbf{r}_1$
$\mathbf{r}_1\mathbf{r}_2$	$\mathbf{r}_1\mathbf{r}_2$	$-\mathbf{r}_2$	$-\mathbf{r}_1$	1

Notice that the even sub algebra of terms generated by scalars and bivectors is isomorphic to the hyperbolic complex numbers $\langle 1, \mathbf{r}_1\mathbf{r}_2 \rangle \simeq \mathbb{H} = \mathbb{R}[j]/(j^2 - 1)$. For vectors $\mathbf{a}_i = x_i\mathbf{r}_1 + y_i\mathbf{r}_2$ and scalar plus bivectors $\mathbf{B}_i = u_i + v_i\mathbf{r}_1\mathbf{r}_2$ we have that

$$\begin{aligned}\mathbf{a}_1\mathbf{a}_2 &= x_1x_2 - y_1y_2 + (x_1y_2 - x_2y_1)\mathbf{r}_1\mathbf{r}_2, \\ \mathbf{a}_1\mathbf{B}_2 &= (x_1u_2 + y_1v_2)\mathbf{r}_1 + (y_1u_2 + x_1v_2)\mathbf{r}_2, \\ \mathbf{B}_1\mathbf{a}_2 &= (u_1x_2 - v_1y_2)\mathbf{r}_1 + (-v_1x_2 + u_1y_2)\mathbf{r}_2, \\ \mathbf{B}_1\mathbf{B}_2 &= u_1u_2 + v_1v_2 + (u_1v_2 + u_2v_1)\mathbf{r}_1\mathbf{r}_2.\end{aligned}$$

The polar coordinates for the red plane are slightly more complicated by the fact that $\cosh(\theta)$ is a strictly positive even function over linearly ordered fields. Since the unit circle of hyperbolic geometry is actually an unbounded hyperbola, we can only parametrize a single branch at a time. For another way to observe the defect in red angles, we can compare their integral definition to that of the integral definition of angles in blue space. The situation is very similar for red angles, with some important differences

$$\theta_R(\mathbf{v}_1, \mathbf{v}_2) = \int_{\gamma} \frac{-ydx + xdy}{x^2 - y^2}.$$

The zero set of the denominator $x^2 - y^2 = (x - y)(x + y) = 0$ in our integrand separates the plane into four quadrants which a path cannot travel between. Therefore we will need to initially restrict our attention to the quadrants $x_i + y_i > 0$, $x_i - y_i > 0$ and $u_i + v_i > 0$, $u_i - v_i > 0$. Since the first de-Rahm cohomology group restricted to any of these quadrants is trivial, we should not and can not normalize our angle. The reader is invited to check that for vectors $\mathbf{v}(\theta) = \cosh(\theta)\mathbf{r}_1 + \sinh(\theta)\mathbf{r}_2$ that $\theta_R(\mathbf{v}(\phi_1), \mathbf{v}(\phi_2)) = \phi_1 - \phi_2$. Hyperbolic magnitudes are also almost the same as their euclidean counterparts, but we need an extra set of absolute values under the square root, so that $|\mathbf{a}| = \sqrt{|\mathbf{a}^2|}$, and $|\mathbf{B}| = \sqrt{|\mathbf{B}\mathbf{B}^\dagger|}$. For brevity (but as a slight abuse of notation) we again say that $v(\theta) = \theta_R(\mathbf{r}_1, \mathbf{v})$. Then for $\mathbf{v} = |\mathbf{v}|[\cosh(v(\theta))\mathbf{r}_1 + \sinh(v(\theta))\mathbf{r}_2]$ and as $\mathbf{B} = |\mathbf{B}|[\cosh(B(\theta)) + \sinh(B(\theta))\mathbf{b}_1\mathbf{b}_2]$ we get

$$\mathbf{a}_1\mathbf{a}_2 = |\mathbf{a}_1||\mathbf{a}_2|[\cosh(a_2(\theta) - a_1(\theta)) + \sinh(a_2(\theta) - a_1(\theta))\mathbf{b}_1\mathbf{b}_2],$$

$$\mathbf{a}_1\mathbf{B}_2 = |\mathbf{a}_1||\mathbf{B}_2|[\cosh(a_1(\theta) - B_2(\theta))\mathbf{b}_1 + \sinh(a_1(\theta) - B_2(\theta))\mathbf{b}_2],$$

$$\mathbf{B}_1\mathbf{a}_2 = |\mathbf{a}_1||\mathbf{B}_2|[\cosh(B_1(\theta) + a_2(\theta))\mathbf{b}_1 + \sinh(B_1(\theta) + a_2(\theta))\mathbf{b}_2],$$

$$\mathbf{B}_1\mathbf{B}_2 = |\mathbf{B}_1||\mathbf{B}_2|[\cosh(B_1(\theta) + B_2(\theta)) + \sinh(B_1(\theta) + B_2(\theta))\mathbf{b}_1\mathbf{b}_2].$$

We can conclude from these formulas that $e^{\theta(\mathbf{r}_1 \wedge \mathbf{r}_2)}\mathbf{v}(x)$ hyperbolically rotates $\mathbf{v}(x)$ counter-clockwise by θ hyperbolic radians, and $\mathbf{v}(x)e^{\theta(\mathbf{r}_1 \wedge \mathbf{r}_2)}$ hyperbolically rotates $\mathbf{v}(x)$ clockwise by θ hyperbolic radians.

Exercise 3 Generalize this discussion to all four quadrants of the x_i, y_i and u_i, v_i planes.

In order to study functions and curves in the red plane we introduce the vector differential operator $\nabla_R = \mathbf{r}_1 \frac{\partial}{\partial x} - \mathbf{r}_2 \frac{\partial}{\partial y}$ called the **red gradient**. Given a potential $f(x, y) = f(\mathbf{x})$ then we can write the tangent line at $\mathbf{r}$ as $[\nabla_R f(\mathbf{r})]^T R(\mathbf{x} - \mathbf{r}) = 0$ where $\mathbf{x} = x\mathbf{r}_1 + y\mathbf{r}_2$ is a vector of variables and

$$R = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

is the metric tensor of our quadratic form. This implies that the red gradient of a potential is red perpendicular to tangent lines of the potential.

For a vector valued function $\mathbf{v}(\mathbf{x}) = p(\mathbf{x})\mathbf{r}_1 + q(\mathbf{x})\mathbf{r}_2$ we have

$$\nabla_R \mathbf{v}(\mathbf{x}) = \frac{\partial p}{\partial x} - \frac{\partial q}{\partial y} + \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y} \right) \mathbf{r}_1 \mathbf{r}_2$$

which is the red divergence of the vector function plus its curl times the unit bivector.

Likewise, in order to understand the effect of the gradient on a scalar plus bivector (hyperbolic complex) valued function, we need to look at a consequence of the chain rule. If a complex expression $u(x, y) + v(x, y)\mathbf{r}_1\mathbf{r}_2$ is a differentiable function of a hyperbolic complex variable $f(z = x + y\mathbf{r}_1\mathbf{r}_2)$ then according to the chain rule $\frac{df}{dx} = \frac{df}{dz} \frac{dz}{dx}$ and $\frac{df}{dy} = \frac{df}{dz} \frac{dz}{dy}$. Solving for $\frac{\partial f}{\partial z}$, setting both expressions equal to each other, and simplifying we get $\frac{df}{dz} \mathbf{r}_1 \mathbf{r}_2 = \frac{df}{dy}$. By substituting in our complex expression for $f(z)$ into this equation, and then comparing the scalar and bivector parts we get two relations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$ called the **hyperbolic Cauchy-Riemann** equations. Now applying the gradient

$$\nabla_R f(z) = (\mathbf{r}_1 \frac{\partial}{\partial x} - \mathbf{r}_2 \frac{\partial}{\partial y})(u(x, y) + v(x, y)\mathbf{r}_1\mathbf{r}_2) = \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) \mathbf{r}_1 + \left(-\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \mathbf{r}_2$$

the hyperbolic Cauchy-Riemann equations tell us $\nabla_R f(z) = 0$.

Exercise 4 Prove that if $f(z) = u(x, y) + v(x, y)\mathbf{r}_1\mathbf{r}_2$ then $\nabla_R u \cdot \nabla_R \mathbf{v} = 0$. For what conditions on $\mathbf{v}(\mathbf{x})$ is $\nabla_R^2 v = 0$? For what conditions on $f(z)$ is $\nabla_R^2 f = 0$?

Exercise 5 Using the previous exercise and the function $f(z) = z^2$, prove that blue $x^2 + y^2 = b$ and green $2xy = g$ circles are red perpendicular wherever they intersect. Prove that if a blue and green circle intersect at a red null point, then their tangent lines at that null point are equal.

Example 3 Almost no one is thinking about the green plane, but I have encountered it in the wild when I did an [independent study with Arthur Baragar](#) on [a paper of E.B. Vinberg](#). It is the GA generated by the basis vectors $\mathbf{g}_1, \mathbf{g}_2$ subject to the relations $\mathbf{g}_1^2 = \mathbf{g}_2^2 = 0$ and $(\mathbf{g}_1 + \mathbf{g}_2)^2 = 2$, which implies $\mathbf{g}_1\mathbf{g}_2 + \mathbf{g}_2\mathbf{g}_1 = 2$ they are not orthonormal as well as $\mathbf{g}_1\mathbf{g}_2 \neq -\mathbf{g}_2\mathbf{g}_1$ our basis vectors do not anti commute. Over a linearly ordered field green geometry is non-definite, because it has a vector $\mathbf{g}_1 + \mathbf{g}_2$ whose square is positive, and a vector $\mathbf{g}_1 - \mathbf{g}_2$ whose square is negative. Green geometry always has visible null vectors, of the forms $k\mathbf{g}_1$ and $k\mathbf{g}_2$, over any field. The multiplication table for the green plane is

Green	1	$\mathbf{g}_1$	$\mathbf{g}_2$	$\mathbf{g}_1 \wedge \mathbf{g}_2$
1	1	$\mathbf{g}_1$	$\mathbf{g}_2$	$\mathbf{g}_1 \wedge \mathbf{g}_2$
$\mathbf{g}_1$	$\mathbf{g}_1$	0	$1 + \mathbf{g}_1 \wedge \mathbf{g}_2$	$-\mathbf{g}_1$
$\mathbf{g}_2$	$\mathbf{g}_2$	$1 - \mathbf{g}_1 \wedge \mathbf{g}_2$	0	$\mathbf{g}_2$
$\mathbf{g}_1 \wedge \mathbf{g}_2$	$\mathbf{g}_1 \wedge \mathbf{g}_2$	$\mathbf{g}_1$	$-\mathbf{g}_2$	1

Notice that the even sub algebra of terms generated by scalars and bivectors is isomorphic to the hyperbolic complex numbers $\langle 1, \mathbf{g}_1 \wedge \mathbf{g}_2 \rangle \simeq \mathbb{H} = \mathbb{R}[j]/(j^2 - 1)$. This is not an accident, because we can identify the green and red plane according to a change of basis using $\mathbf{g}_1 = \frac{1}{\sqrt{2}}(\mathbf{r}_1 + \mathbf{r}_2)$ and $\mathbf{g}_2 = \frac{1}{\sqrt{2}}(\mathbf{r}_1 - \mathbf{r}_2)$. This does not mean that green geometry is not worth studying. From a blue perspective, red and green geometry are related by a simple rotation. Is it true that from the perspective of red geometry, the green and blue planes are identifiable by a red rotation? Or from green geometry, red and blue planes are identifiable by a green rotation?

Returning back to the usual flow of the narrative, for vectors $\mathbf{a}_i = x_i \mathbf{g}_1 + y_i \mathbf{g}_2$ and scalar plus bivectors $\mathbf{B}_i = u_i + v_i \mathbf{g}_1 \wedge \mathbf{g}_2$ we have that

$$\mathbf{a}_1 \mathbf{a}_2 = x_1 y_2 + y_1 x_2 + (x_1 y_2 - x_2 y_1) \mathbf{g}_1 \wedge \mathbf{g}_2,$$

$$\mathbf{a}_1 \mathbf{B}_2 = (x_1 u_2 - x_1 v_2) \mathbf{g}_1 + (y_1 u_2 + y_1 v_2) \mathbf{g}_2,$$

$$\mathbf{B}_1 \mathbf{a}_2 = (u_1 x_1 + v_2 x_1) \mathbf{g}_1 + (u_1 y_2 - v_1 y_2) \mathbf{g}_2,$$

$$\mathbf{B}_1 \mathbf{B}_2 = u_1 u_2 + v_1 v_2 + (u_1 v_2 + u_2 v_1) \mathbf{g}_1 \wedge \mathbf{g}_2.$$

Exercise 6 Show that $\mathbf{v}(\theta) = (\frac{e^{-\theta}}{\sqrt{2}}, \frac{e^{\theta}}{\sqrt{2}})$ is a counterclockwise constant green velocity parametrization for the positive branch of the green unit circle $2xy = 1$. Define the green angle in the first quadrant $x, y > 0$ of the plane by

$$\theta_G(\mathbf{v}_1, \mathbf{v}_2) = \int_{\gamma} \frac{-y dx + x dy}{2xy},$$

where gamma is a path from the tip of $\mathbf{v}_1$ to the tip of $\mathbf{v}_2$ which stays in the first quadrant. Show for vectors $\mathbf{v}(\theta) = \frac{1}{\sqrt{2}}(e^{-\theta} \mathbf{g}_1 + e^{\theta} \mathbf{g}_2)$ that $\theta_G(\mathbf{g}_1 + \mathbf{g}_2, \mathbf{v}(\phi)) = \phi$ then generalize this discussion to all four quadrants.

Exercise 7* Write down and interpret the usual polar forms of vector / scalar plus bivector multiplications. Interpret geometrically what is happening.

Exercise 8 Define the **green gradient** as $\nabla_G = \mathbf{g}_1 \frac{\partial}{\partial y} + \mathbf{g}_2 \frac{\partial}{\partial x}$. Show that the green gradient of a vector valued function is its green divergence plus its curl times $\mathbf{g}_1 \wedge \mathbf{g}_2$.

Exercise 9* Show that a green complex expression $u(x, y) + v(x, y) \mathbf{g}_1 \wedge \mathbf{g}_2$ is a differentiable function of a complex variable exactly when $\nabla_G(u(x, y) + v(x, y) \mathbf{g}_1 \wedge \mathbf{g}_2) = 2 \frac{\partial u}{\partial x} \mathbf{g}_2$. (??)

* Surely I am making some basic mistake here, neither of these are what I would expect from a GA which is a change of basis away from red geometry.

If you made it this far, congratulations. There is still so much to say, but this paper is getting too long, so Ill stop here.